

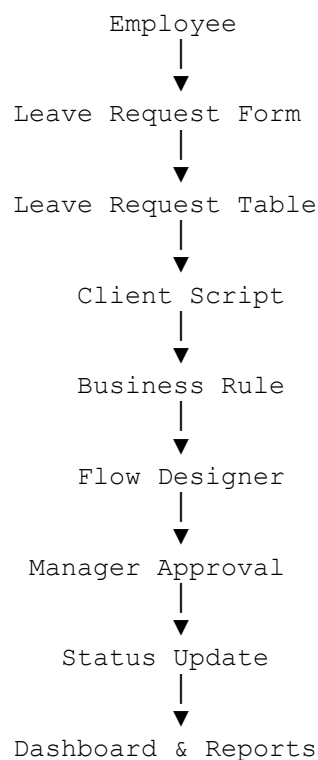
PROJECT DESIGN PHASE

Smart Leave Management System

2. System Architecture Design

The application follows a layered architecture where employees submit leave requests through a custom form, requests are stored in a custom table, and approvals are managed through Flow Designer. Business Rules automate leave duration calculation, while Client Scripts perform form validation.

System Architecture Diagram



Architecture Components

Employee

Submits leave requests.

Leave Request Form

Provides the leave request submission interface.

Leave Request Table

Stores leave request information.

Client Script

Validates leave dates before submission.

Business Rule

Calculates total leave days automatically.

Flow Designer

Automates approval workflows.

Dashboard

Displays request statistics and reports.

3. Database Design

A custom table was designed to store all employee leave request information.

Table Name

Leave Request

Table Fields

Leave Request
PK : Request Number
Employee Name Leave Type Start Date End Date Total Days Reason Approval Status

Field Name | Data Type | Description

Employee Name | String | Employee name

Leave Type | Choice | Type of leave

Start Date | Date | Leave start date

End Date | Date | Leave end date

Total Days | Integer | Calculated leave days

Reason | String | Leave reason

Approval Status | Choice | Approval state

4. Form Design

A structured form was designed to collect leave request information.

Form Layout

Section 1: Employee Details

Employee Name

Section 2: Leave Information

Leave Type

Start Date

End Date

Reason

Form Design Diagram

Leave Request Form
Employee Details
Employee Name
Leave Details
Leave Type
Start Date
End Date
Total Days
Reason
Submit

Design Considerations

- Easy navigation
- Mandatory field validation

- Reduced user errors
- User-friendly interface

5. Application Design

A custom ServiceNow application was designed to provide self-service leave request management.

Application Name

Smart Leave Management System

Form Fields

Field Name | Type

Employee Name | Single Line Text

Leave Type | Choice

Start Date | Date

End Date | Date

Total Days | Integer

Reason | Multi Line Text

Approval Status | Choice

6. Workflow Design

The workflow was designed using ServiceNow Flow Designer.

Workflow Diagram

Leave Request Created

↓

Set Status = Pending Approval

↓

Ask For Approval

↓

If Approved

└─ Yes → Update Status = Approved

└─ No → Update Status = Rejected

Workflow Components

Trigger

Leave Request Created

Approval Action

Routes leave request for manager approval.

Business Rule

Calculates total leave days automatically.

Client Script

Validates start date and end date.

IF Condition

Checks approval status.

Status Update

Updates request status after approval decision.

Dashboard Reporting

Displays total, approved, rejected, and pending requests.

11. Design Advantages

Scalability

Supports increasing leave request volume.

Maintainability

Components can be modified independently.

Automation

Reduces manual effort through workflows and business rules.

User Experience

Simple and intuitive interface.

Transparency

Provides visibility into leave request status.

Accuracy

Automatic leave calculation reduces errors.

Conclusion

The Project Design Phase defined the complete structure of the Smart Leave Management System. The architecture, database design, form design, workflow design, validation logic, and reporting dashboard provide a strong foundation for implementation. The design ensures efficient leave request handling, approval automation, accurate leave calculation, and real-time request tracking while maintaining scalability and usability.